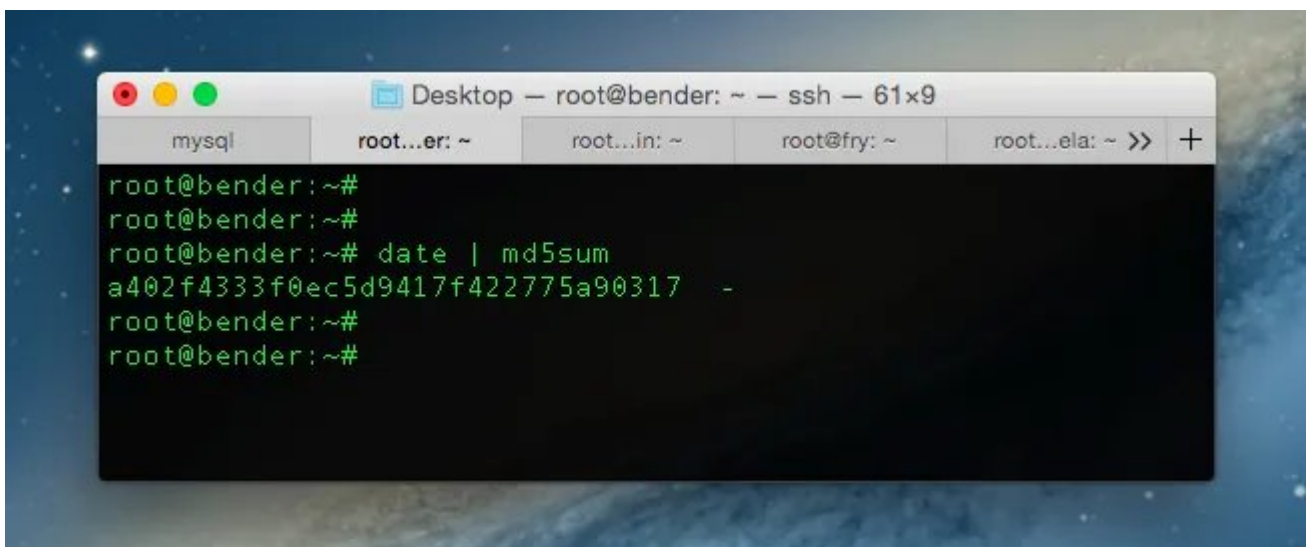


10 Ways to Generate a Random Password from the Linux Command Line

BY **LOWELL HEDDINGS** PUBLISHED DEC 20, 2014

One of the great things about Linux is that you can do the same thing hundreds of different ways—even something as simple as generating a random pas



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One of the great things about Linux is that you can do the same thing hundreds of different ways—even something as simple as generating a random password can be accomplished with dozens of different commands. Here's 10 ways you can do it.

We gathered all of these commands from [Command-Line Fu](#) and tested them out on our own Linux PC to make sure they work. You should be able to use at least some of these on Windows with [Cygwin](#). [Link copied to clipboard](#) You don't test all of them--the last one definitely works though.

Generate a Random Password

For any of these random password commands, you can either modify them to output a different password length, or you can just use the first x characters of the generated password if you don't want such a long password. Hopefully you're using a password manager like [LastPass](#) anyway so you don't need to memorize them.

This method uses SHA to hash the date, runs through base64, and then outputs the top 32 characters.

```
date +%s | sha256sum | base64 | head -c 32 ; echo
```

This method used the built-in `/dev/urandom` feature, and filters out only characters that you would normally use in a password. Then it outputs the top 32.

```
< /dev/urandom tr -dc _A-Z-a-z-0-9 | head -c${1:-32};echo;
```

This one uses openssl's `rand` function, which may not be installed on your system. Good thing there's lots of other examples, right?

```
openssl rand -base64 32
```

This one works a lot like the other urandom one, but just does the work in reverse. Bash is very powerful!

```
tr -cd '[:alnum:]' < /dev/urandom | fold -w30 | head -n1
```

Here's another example that filters using the strings command, which outputs printable strings from a file, which in this case is the urandom feature.

```
strings /dev/urandom | grep -o '[:alnum:]' | head -n 30 | tr -d '\n'; echo
```

Here's an even simpler version of the urandom one.

```
< /dev/urandom tr -dc _A-Z-a-z-0-9 | head -c6
```

This one manages to use the very useful dd command.

```
dd if=/dev/urandom bs=1 count=32 2>/dev/null | base64 -w 0 | rev | cut -b 2- | rev
```

You can even create a random left-hand password, which would let you type your password with one hand.

```
</dev/urandom tr -dc '12345!@#%qwertQWERTasdfgASDFGzxcvbZXCVB' | head -c8; echo ""
```

If you're going to be using this all the time, it's probably a better idea to put it into a function. In this case, once you run the command once, you'll be able to use `randpw` anytime you want to generate a random password. You'd probably want to put this into your `~/.bashrc` file.

```
randpw(){ < /dev/urandom tr -dc _A-Z-a-z-0-9 | head -c${1:-16};echo;}
```

You can use this same syntax to make any of these into a function—just replace everything inside the `{ }`

And here's the easiest way to make a password from the command line, which works in Linux, Windows with Cygwin, and probably Mac OS X. I'm sure that some people will complain that it's not as random as some of the other options, but honestly, it's random enough if you're going to be using the whole thing.

Yeah, that's even easy enough to remember.

There's loads of other ways that you can create a random password from the command line in Linux—for instance, the `mkpasswd` command, which can actually assign the password to a Linux user account. So what's your favorite way?

	Linux Commands
Files	tar · pv · cat · tac · chmod · grep · diff · sed · ar · man · pushd · popd · fsck · testdisk · seq · fd · pandoc · cd · \$PATH · awk · join · jq · fold · uniq · journalctl · tail · stat · ls · fstab · echo · less · chgrp · chown · rev · look · strings · type · rename · zip · unzip · mount · umount · install · fdisk · mkfs · rm · rmdir · rsync · df · gpg · vi · nano · mkdir · du · ln · patch · convert · rclone · shred · srm · scp · gzip · chattr · cut · find · umask · wc · tr
Processes	alias · screen · top · nice · renice · progress · strace · systemd · tmux · chsh · history · at · batch · free · which · dmesg · chfn · usermod · ps · chroot · xargs · tty · pinky · lsof · vmstat · timeout · wall · yes · kill · sleep · sudo · su · time · groupadd · usermod · groups · lshw · shutdown · reboot · halt · poweroff · passwd · lscpu · crontab · date · bg · fg · pidof · nohup · pmap
Networking	netstat · ping · traceroute · ip · ss · whois · fail2ban · bmon · dig · finger · nmap · ftp · curl · wget · who · whoami · w · iptables · ssh-keygen · ufw · arping · firewalld

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Lowell Heddings

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Lowell Heddings is the founder of How-To Geek. Over the last decade, Lowell has personally written more than 1000 articles which have been viewed by over 250 million people....

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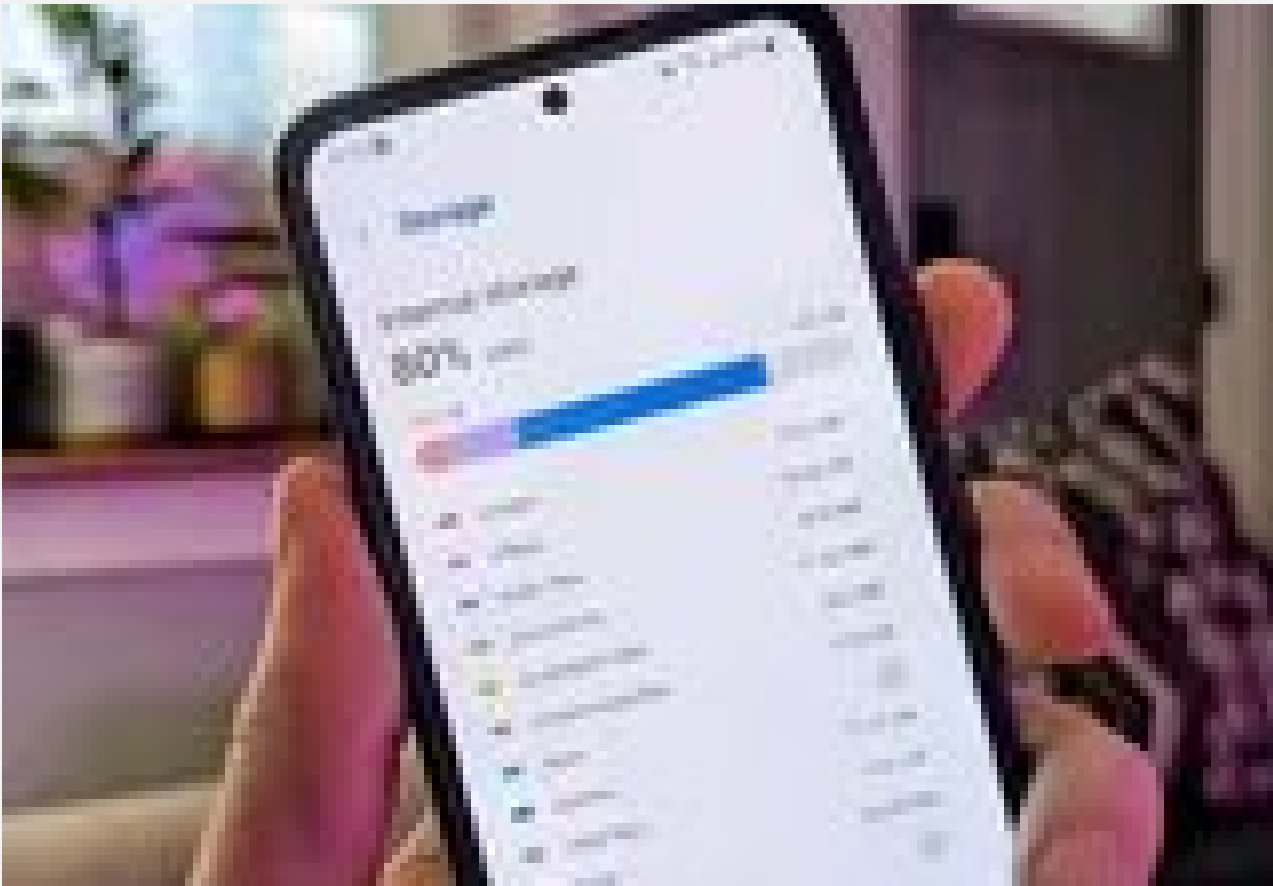
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